

Objective:

A good way to study a system, natural or engineered, is to find its invariances because these quantities define or label the system completely. An invariant quantity is related to a symmetry operation on the system. Since the set of symmetry operations form a Group, it becomes imperative for scientists and engineers to understand the basic ideas of Group and its representations in vector spaces. This brings in the need to emphasize on the structural aspects of algebra rather than only computational. The codes in Information and Communication Technology are based on certain discrete groups and Fields. Hence the objective of the course is to give a comprehensive understanding of these algebraic structures. Emphasis is on the similarity of the various structures, i.e, homomorphism so that students learn to map seemingly different problems using the underlying abstract algebraic structure.

Course Outline

1. Groups:
Basic Definitions, Subgroups, Cosets, Normal Subgroups, Lagrange's Theorem, Isomorphism, Homomorphism, Quotient groups, Cayley's Theorem.
2. Rings:
Definition, types of rings , zero divisors , integral domains, Fields, Characteristic of a field.
3. The number system:
Examples of Rings and Fields in the number systems.
4. Polynomials:
Definition, The division Algorithm, Factorization.
5. Quotient Rings:
Homomorphisms, Ideals, Quotient rings, Quotient rings of Polynomial Rings, Factorization and Ideals.
6. Field Extensions:
Simple Extensions, Degree of Extension, Splitting Fields, Finite Fields.

Books

1. Topics in Algebra
I.N. Herstein
2. Abstract Algebra, An Introduction
Thomas W. Hungerford
3. Algebra
Michael Artin

Grading

Two midsem test 20%+ 20% and an endsem 40%.

If possible Quiz will be conducted.

Quiz and attendance will comprise 20% of evaluation process.

Outcome:

Students learn the important algebraic structures of Groups and Fields which one encounters in various areas of Science and Engineering. They learn homomorphism and the important theorems of homomorphism which help in understanding and interpreting various aspects of a given algebraic structure. Introduction to a variety of Rings and Fields help them to generalize the structure of integers, real and complex numbers. Some of these fields like the $\langle \mathbb{Z}_n, \oplus \rangle$ and $\langle \mathbb{Z}_n^*, \otimes \rangle$ are useful in coding and cryptography .